

Image Compression Using SVD

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Definition. Let A be an $m \times n$ matrix. A matrix norm $\|A\|$ is a number assigned to the matrix A such that

1. $\|A\| \geq 0$ and $\|A\| = 0$ if and only if A is the zero matrix
2. $\|\alpha A\| = |\alpha| \|A\|$ for any scalar α
3. $\|A + B\| \leq \|A\| + \|B\|$

Definition. The Frobenius norm for a matrix A is defined by

$$\|A\|_F = \sqrt{\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2}$$

1. Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ -3 & 4 & 0 \end{bmatrix}$.

Proof. $\|A\|_F = \sqrt{1+4+1+1+4+9+4} = \sqrt{24} = 2\sqrt{6}$. □

2. Show that $\|I_n\|_F = \sqrt{n}$, where I_n is the $n \times n$ identity matrix.

Proof. $\|I_n\|_F = \sqrt{1^2 \cdot n} = \sqrt{1 \cdot n} = \sqrt{n}$. □

3. Show that $\|A\|_F$ is a matrix norm.

Proof. $\|A\|_F \geq 0$ because the root of a sum of non-negatives is non-negative.

$$\|A\|_F = 0 \iff \text{the entries equal zero.}$$

$$\|\alpha A\|_F = \sqrt{\sum_{j=1}^n \sum_{i=1}^m |\alpha a_{ij}|^2} = \sqrt{|\alpha|^2} \sqrt{\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2} = |\alpha| \|A\|_F.$$

$$\|A + B\|_F^2 = \sum_{j=1}^n \sum_{i=1}^m |a_{ij} + b_{ij}|^2 \leq \sum_{j=1}^n \sum_{i=1}^m (|a_{ij}| + |b_{ij}|)^2 =$$

$$\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2 + 2 \sum_{j=1}^n \sum_{i=1}^m |a_{ij}| |b_{ij}| + \sum_{j=1}^n \sum_{i=1}^m |b_{ij}|^2 \leq$$

$$\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2 + 2\sqrt{\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2} \sqrt{\sum_{j=1}^n \sum_{i=1}^m |b_{ij}|^2} + \sum_{j=1}^n \sum_{i=1}^m |b_{ij}|^2$$

[Cauchy-Schwarz Inequality].

$$\|A + B\|_F^2 \leq \|A\|_F^2 + 2\|A\|_F \|B\|_F + \|B\|_F^2.$$

$$\therefore \|A + B\|_F \leq \|A\|_F + \|B\|_F. \quad \square$$

4. Show that $\|A\|_F^2 = \text{trace}(A^T A)$, where $\text{trace}(A)$ is the sum of diagonal entries of A .

Proof. Let $B = A^T A$.

$$b_{jj} = (\text{row}_j \text{ of } A^T) \cdot (\text{col}_j \text{ of } A) = \sum_{i=1}^m (a_{ij})(a_{ij}) = \sum_{i=1}^m a_{ij}^2.$$

$$\therefore \text{trace}(B) = \sum_{j=1}^n b_{jj} = \sum_{j=1}^n \left(\sum_{i=1}^m a_{ij}^2 \right) = \|A\|_F^2. \quad \square$$

- 5a. Show that $\text{trace}(BA) = \text{trace}(AB)$.

Proof. $(AB)_{ii} = \sum_j a_{ij} b_{ji}$.

$$\text{trace}(AB) = \sum_i (ab)_{ii} = \sum_i \sum_j a_{ij} b_{ji}.$$

$$(BA)_{jj} = \sum_i b_{ji} a_{ij}.$$

$$\text{trace}(BA) = \sum_j (ba)_{jj} = \sum_j \sum_i b_{ji} a_{ij}.$$

$$\therefore \text{trace}(BA) = \text{trace}(AB). \quad \square$$

- b. Use (a) to show that $\|AO\|_F = \|A\|_F$ for any orthogonal matrix O .

Proof. $\|AO\|_F^2 = \text{trace}((AO)^T(AO)) = \text{trace}(O^T A^T AO) = \text{trace}(AOO^T A^T)$

[part (a)].

$$= \text{trace}(AIA^T) = \text{trace}(AA^T) = \text{trace}(A^T A) = \|A\|_F^2.$$

$$\therefore \|AO\|_F = \|A\|_F. \quad \square$$

- c. Use (b) to show that $\|A\|_F^2 = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2$.

Proof. Let $A = U\Sigma V^T$

[Singular Value Decomposition].

$$\|A\|_F^2 = \|U\Sigma V^T\|_F^2 = \|\Sigma\|_F^2 = \sum_{j=1}^n \sum_{i=1}^m |\sigma_{ij}|^2 = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2. \quad \square$$

- 6a. Prove that if the columns of B are linearly independent, then so are the columns of AB .

Proof. Suppose $(AB)\vec{x} = \vec{0}$.

$$A(B\vec{x}) = \vec{0}.$$

Since A is square and linearly independent, it is invertible with $B\vec{x} = \vec{0}$.

Also, since B is linearly independent, the equation $B\vec{x} = \vec{0}$ has only the trivial solution, $\vec{x} = \vec{0}$.

$\therefore$ The columns of AB must be linearly independent. □

b. Prove that $\text{rank}(AB) \leq \text{rank}(B)$

Proof. Let $\vec{x}$ be in the null space of B .

$$AB\vec{x} = A(B\vec{x}) = A(\vec{0}) = \vec{0}.$$

Hence, $\text{Nul}(B) \subseteq \text{Nul}(AB)$.

$$\therefore \text{nullity}(AB) \geq \text{nullity}(B) \implies \text{rank}(AB) \leq \text{rank}(B). \quad \square$$

c. Prove that if U is invertible, then $\text{rank}(UA) = \text{rank}(A)$.

Proof. Let U be an invertible matrix.

$$\text{rank}(UA) \leq \text{rank}(A)$$

[part (b)].

$$\text{rank}(A) = \text{rank}(U^{-1}(UA)) \leq \text{rank}(UA)$$

[part (b)].

$$\therefore \text{rank}(UA) = \text{rank}(A). \quad \square$$

d. Prove that if V is invertible, then $\text{rank}(AV) = \text{rank}(A)$.

Proof. Let V be an invertible matrix.

$$\text{rank}((AV)^T) = \text{rank}(V^T A^T) = \text{rank}(A^T)$$

[part (c)].

$$\therefore \text{rank}(AV) = \text{rank}(A). \quad \square$$

7a. Let $A = U\Sigma V^T$ be a singular value decomposition of an $m \times n$ matrix A . Let $\sigma_1 \geq \sigma_2 \geq \dots \sigma_r$ be all the nonzero singular values of A . Prove that the rank of A is r .

Proof. Let $A = U\Sigma V^T$ be a singular value decomposition of an $m \times n$ matrix A , where

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_r & \\ & & & & 0 \end{bmatrix}, \text{ with singular values satisfying } \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0, \text{ and all remaining diagonal}$$

entries equal to 0. Since U and V are orthogonal matrices, they are invertible. Since multiplication by an invertible matrix does not change rank, $\text{rank}(A) = \text{rank}(U\Sigma V^T) = \text{rank}(\Sigma)$. Thus it suffices to compute the rank of Σ . The matrix Σ is diagonal, and its only nonzero entries are the singular values $\sigma_1, \dots, \sigma_r$. Therefore Σ has exactly r nonzero diagonal entries. For a diagonal matrix, the rank equals the number of nonzero diagonal entries, since each nonzero diagonal entry contributes one pivot. Hence, $\text{rank}(\Sigma) = r$. Therefore, $\text{rank}(A) = r$. $\square$

b. Let $A = U\Sigma V^T$ be a singular value decomposition of an $m \times n$ matrix A . Let $k \leq r = \text{rank}(A)$. Define $A_k = U\Sigma_k V^T$, where $\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0)_{m \times n}$ where $\sigma_1 \geq \sigma_2 \geq \dots \sigma_k > 0$. Prove that the rank of A_k is k .

Proof. Let $A = U\Sigma V^T$ be a singular value decomposition of an $m \times n$ matrix A , and let $A_k = U\Sigma_k V^T$, where $\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0)_{m \times n}$, with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$. Since U and V are orthogonal matrices, they are invertible. Multiplication by invertible matrices does not change rank, so $\text{rank}(A_k) = \text{rank}(U\Sigma_k V^T) = \text{rank}(\Sigma_k)$. The matrix Σ_k is diagonal and has exactly k nonzero diagonal entries, namely $\sigma_1, \dots, \sigma_k$. For a diagonal matrix, the rank equals the number of nonzero diagonal entries. Therefore, $\text{rank}(\Sigma_k) = k$. Hence, $\text{rank}(A_k) = k$. $\square$

8. Consider the outer product form $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$.

a. Suppose we have a grayscale picture that is 3072×2049 pixels in size. Each picture is one of 256 shades of gray represented by a number between 0 and 255. Assuming this information is stored in a matrix and we wanted to send it to someone digitally, how many numbers would have to be transmitted to share all of the information?

Since the image is represented by a 3072×2049 matrix, the total number of entries is

$$3072 \cdot 2049 = 6,294,528.$$

Thus, 6,294,528 numbers must be transmitted.

b. To reduce the amount of information that is sent, suppose we instead sent a rank 200 approximation. What information would have to be sent and how many numbers would need to be sent?

A rank 200 approximation has the form

$$A_{200} = \sum_{i=1}^{200} \sigma_i u_i v_i^T.$$

To transmit this approximation, we must send:

- 200 singular values,
- 200 left singular vectors in $\mathbb{R}^{3072}$,
- 200 right singular vectors in $\mathbb{R}^{2049}$.

Hence the total number of transmitted numbers is

$$200 + 200(3072) + 200(2049) = 200 + 614,400 + 409,800 = 1,024,400.$$

Thus, 1,024,400 numbers are required.

c. Why do the smallest singular values contribute very little to the image?

The singular values measure the contribution of each term

$$\sigma_i u_i v_i^T$$

to the image. Small singular values correspond to terms that contribute very little, usually representing fine details or noise. Therefore, removing the smallest singular values changes the image only slightly while significantly reducing the amount of data stored or transmitted.

d. MATLAB section

ii. Based on the images, about how many singular values should be kept to retain approximately 75% of the information contained in the Eiffel Tower?

Based on the cumulative percentage plot, approximately the first r singular values where the curve reaches 0.75 should be retained.